Towards Trustworthy Network Security: Evaluating Concept-Based Interpretability and Neuro-Symbolic Out-of-Distribution Detection

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***Abstract*—Machine learning (ML)-based network intrusion detection systems (NIDS) face two persistent deployment challenges: lack of interpretability, and poor handling of novel attack families unseen during training (the open-set problem). We present a joint evaluation of two complementary interpretable architectures applied to IoT traffic classification and open-set detection. First, Concept Bottleneck Models (CBMs) route predictions through a layer of human-defined traffic concepts and support test-time intervention. Second, Neuro-Symbolic NIDS (NeSy-NIDS) encodes domain-expert attack signatures as differentiable threshold rules with learnable parameters, yielding an exactly binary rule-activation space via k-annealing and the Straight-Through Estimator. NeSy-NIDS additionally incorporates a neural fallback gated by a learnable mixing coefficient α. Both architectures independently use Mahalanobis distance in their respective representation spaces for open-set scoring. On CTU-IoT-23 and CIC-IoT-2023, neither method incurs any classification accuracy cost (F1 within ± 0.004 of an unconstrained MLP). On CTU-IoT-23, rule-space Mahalanobis achieves OOD AUROC = 0.909 (σ = 0.017), comparable to the unconstrained MLP baseline (0.913). On CIC-IoT-2023, OOD performance depends critically on how tightly bottleneck training is coupled to the classification objective. SequentialCBM, which learns concepts independently of the task head, achieves AUROC = 0.839—exceeding the Decision Tree baseline (0.809)—while JointCBM degrades monotonically from 0.627 (γ=1.0) to 0.452 (γ=0.1) and NeSy-NIDS reaches 0.613. We term this the *task-coupling OOD penalty*: jointly optimising the bottleneck for classification collapses the distributional spread required for Mahalanobis OOD separation. NeSy-NIDS additionally provides semantically consistent threshold drift and a controllable interpretability-vs-OOD trade-off via α-regularisation, at zero F1 cost.**

***Index Terms*—IoT network intrusion detection, concept bottleneck models, differentiable rule learning, open-set detection, out-of-distribution detection, interpretable machine learning**

I. Introduction

Internet of Things (IoT) devices now constitute a dominant fraction of network endpoints. Their minimal patch cycles and heterogeneous firmware make them attractive targets for botnet recruitment, distributed denial-of-service (DDoS) amplification, and command-and-control (C&C) infrastructure. Network-level intrusion detection systems (NIDS) are a practical first line of defence: passive on traffic flows, requiring no modification to end-devices.

Deep learning (DL)-based NIDS achieve strong classification accuracy for known attacks; however, two structural limitations impede operational deployment. First, the *interpretability gap*: DL models provide no human-auditable justification for flagged alerts. Without knowing *which* traffic features triggered a decision, operators cannot validate alerts, tune policies, or satisfy regulatory requirements [1], [2]. Post-hoc explainers address this symptom but operate outside the model; even the most capable post-hoc NIDS explainer (xNIDS [3]) achieves its fidelity gains only relative to LIME and SHAP, while remaining external to the model's decision process.

Second, the *open-set problem*: NIDS trained on a closed set of known attacks will inevitably encounter novel families during deployment [2]. A model that confidently assigns novel traffic to a known class provides false assurance. Existing approaches either retrain periodically or deploy a secondary anomaly detector, neither of which uses the model's own representations for principled open-set scoring [4].

In this paper, we tackle both problems by independently embedding interpretability *into* each model's architecture and evaluating its effect on open-set detection. Two complementary architectures are studied: (i) **CBMs** [5]: predictions are forced through an intermediate layer of human-defined concepts (e.g., *is_high_rate*, *is_beaconing*), making each intermediate decision directly auditable and enabling test-time interventions; (ii) **NeSy-NIDS**: domain-expert attack signatures are encoded as parameterised threshold rules whose thresholds are jointly learned via k-annealing and the Straight-Through Estimator (STE) [6], [7], producing an exactly binary rule-activation vector at inference.

Unlike xNIDS [3], which generates post-hoc explanations from a black-box model, both architectures are inherently interpretable: intermediate representations consist of human-defined concepts (CBM) or exact binary rules (NeSy-NIDS). Unlike OWAD [4], which detects normality shift in deployed detectors over time, we address unknown-class detection at single-inference granularity in models whose representation spaces are human-interpretable.

Contributions:

1) **Task-coupling OOD penalty**: we identify and explain a failure mode specific to jointly trained bottleneck architectures on CIC-IoT-2023. OOD AUROC degrades monotonically with task-coupling strength (Sequential-CBM: 0.839 > DT: 0.809 > JointCBM γ=1.0: 0.627 > NeSy-NIDS: 0.613 > JointCBM γ=0.1: 0.452), demonstrating that joint optimisation for classification—not compression per se—destroys the distributional spread required for Mahalanobis OOD separation.
2) **Zero accuracy cost**: the first joint evaluation of CBMs and differentiable rule learning for IoT NIDS under open-set evaluation conditions, validating that neither architecture sacrifices classification accuracy relative to an unconstrained MLP (F1 within ± 0.004 across both datasets).
3) **100% rule crispness**: NeSy-NIDS with k-annealing and STE produces strictly binary rule activations at inference without a post-quantisation step, yielding syntactically verifiable if-then rules for deployment audit.
4) **Deployable interpretability trade-off**: a controllable α-gate provides a Pareto front between rule reliance and OOD detectability, selectable at inference time without full retraining.
5) **Concept leakage quantification**: a γ-fidelity regulariser eliminates concept leakage in CBMs, simultaneously improving OOD AUROC on CTU-IoT-23 from 0.812 (γ=0) to 0.884 (γ=0.5) at constant classification F1.

Both models are trained on the publicly labelled CTU-IoT-23 [8] and CIC-IoT-2023 [9] datasets, containing 4/9 and 5/29 known/unknown attack families, respectively.

II. Related Work

A. Interpretability in DL-based NIDS

Post-hoc explanation techniques such as feature attribution [3] and concept drift explanation [10] help operators understand existing DL-NIDS, but do not constrain the model's behaviour. Wei et al. [3] (xNIDS) demonstrate that sparse group lasso over groups of network features outperforms LIME and SHAP on fidelity, stability, and completeness, and their technique outputs concrete firewall rules. Han et al. [11] (DeepAID) formulate explanation constraints as optimisation problems for unsupervised DL-based security anomaly detectors. Han et al. [12] (GEAD) derive global rules for DL-based anomaly detectors via decision regression trees. All these approaches are *post-hoc*, leaving the underlying model opaque. Our work differs by making each model inherently interpretable, so that explanations are guaranteed faithful by construction.

B. Concept Bottleneck Models

Koh et al. [5] introduced CBMs as a two-stage architecture in which predictions are forced through an intermediate layer of human-defined concepts, enabling test-time intervention. Zarlenga et al. [13] propose Concept Embedding Models (CEMs) that extend CBMs to higher-dimensional concept representations without sacrificing intervention capability. Yuksekgonul et al. [14] show that any pre-trained network can be retroactively converted to a CBM via post-hoc concept alignment. No prior work applies CBMs in the NIDS domain; our paper provides the first such analysis, including the first quantitative study of concept leakage in network traffic classification.

C. Differentiable and Neuro-Symbolic Rule Learning

Petersen et al. [15] train deep networks of binary logic gates end-to-end via continuous relaxation and discretisation, demonstrating that binary differentiable architectures match the accuracy of standard neural networks. Yang et al. [16] show that the STE can enforce discrete logical constraints in neural training, providing gradient-based learning of Boolean conditions. Pryor et al. [17] (NeuPSL) jointly learn neural and probabilistic soft logic parameters. Cheng et al. [18] learn compositional logical rules for knowledge graph reasoning. NeSy-NIDS adapts this family of neuro-symbolic models to IoT network traffic by representing logical rules as conjunctions of thresholded features with learnable bounds, enabling direct interpretation by security analysts.

D. Open-Set and OOD Detection in Security

Lee et al. [19] establish Mahalanobis distance over the penultimate feature space as a principled OOD detector superior to maximum-softmax [20]. Fort et al. [21] show that representation quality dominates OOD detection performance; stronger encoders make Mahalanobis distance dramatically more effective. Han et al. [4] (OWAD) address gradual concept drift in deployed NIDS through hypothesis testing on output probability distributions. Yang et al. [10] (CADE) diagnose concept drift via contrastive analysis of security ML representations. Yang et al. [22] adapt lightweight open-set detection to Industrial IoT (IIoT) network anomaly detection. Mirsky et al. [23] (Kitsune) detect intrusions via an online ensemble of autoencoders fitted incrementally to raw packet statistics; while effective for anomaly scoring, the internal feature representations are not human-interpretable and Kitsune provides no mechanism for test-time expert intervention. Unlike OWAD, CADE, and Kitsune, our work focuses on learning inherently interpretable models and analysing how representation-space compression affects OOD detectability at single-inference granularity.

E. ML Pitfalls and IoT Datasets

Arp et al. [2] document ten pitfalls in deploying ML for security, including unrealistic closed-world evaluation and lack of explainability, directly motivating our interpretable, open-set evaluation framework. De Keersmaeker et al. [24] survey more than 70 public IoT network traffic datasets, identifying CTU-IoT-23 and CIC-IoT-2023 as two of the most comprehensive for ML-based NIDS research. Alotaibi and Maffeis [25] (Mateen) achieve state-of-the-art online network anomaly detection using autoencoder ensembles.

III. Methodology

Fig. 1 illustrates the two architectures evaluated in this work.

A. Datasets

CTU-IoT-23 [8] provides labelled network flows from IoT devices captured under controlled laboratory conditions. We use a 37-feature version pre-processed from Zeek logs (packet/byte counts, inter-arrival times, flag counts). Four classes are treated as *known* (Benign, C&C-HeartBeat, DDoS, Okiru); nine attack families are held out as *unknown* for OOD evaluation.

CIC-IoT-2023 [9] covers 33 attack scenarios across 105 IoT devices. After stratified subsampling of the ≈40M-row dataset, the feature set comprises 39 z-score normalised columns. Five classes are known; 29 form the unknown set. Data splits follow a 70/15/15 stratified partition for both datasets.

B. Concept Bottleneck Models

A CBM [5] factors the classifier $f\colon \mathbf{x} \mapsto y$ into two stages:

$$\hat{\mathbf{c}} = g(\mathbf{x}), \quad \hat{y} = h(\hat{\mathbf{c}}) = f(\mathbf{x}), \tag{1}$$

where $g\colon \mathbb{R}^d \to [0,1]^K$ is the concept encoder (d is the number of raw input features: 37 for CTU-IoT-23, 39 for CIC-IoT-2023), $h\colon [0,1]^K \to \mathbb{R}^C$ is the classifier (C is the number of known classes: 4 for CTU-IoT-23, 5 for CIC-IoT-2023), and the equality $h(\hat{\mathbf{c}}) = f(\mathbf{x})$ makes the factorisation $f = h \circ g$ explicit. We define $K{=}8$ domain-derived flow concepts per dataset (e.g., *is_high_rate*, *is_beaconing*, *is_incomplete_handshake*), approximated as soft binary targets from threshold rules on raw features.

Training variants. *JointCBM* ($\gamma{=}0$): standard joint training minimising cross-entropy and binary concept losses simultaneously. *JointCBM* ($\gamma{=}0.5$): joint training augmented with a concept-fidelity penalty:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda\,\mathcal{L}_{\mathrm{BCE}} + \gamma\,\|\hat{\mathbf{c}} - \mathbf{c}_{\mathrm{gt}}\|^2, \tag{2}$$

where $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss on class predictions, $\mathcal{L}_{\mathrm{BCE}}$ is the binary cross-entropy loss on concept predictions, $\mathbf{c}_{\mathrm{gt}} \in \{0,1\}^K$ are the ground-truth binary concept labels, $\lambda{\geq}0$ weights the concept supervision, and $\gamma{\geq}0$ is the fidelity penalty strength. The term $\gamma\,\|\hat{\mathbf{c}} - \mathbf{c}_{\mathrm{gt}}\|^2$ penalises concept activations that deviate from their ground-truth binary targets, mitigating *concept leakage*—the encoding of class-discriminative information beyond the declared concept layer. *SequentialCBM*: g is trained and frozen first; h is then trained on $\hat{\mathbf{c}}$, decoupling concept learning from classification. *HybridCBM*: extends JointCBM with a skip connection—the encoder embedding is appended to the concept predictions before the classifier—allowing the model to leverage latent structure beyond the declared concept vocabulary while retaining interpretability at the concept layer.

OOD scoring. Concept activation vectors from training samples are used to fit per-class Gaussians $\mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma})$ with shared regularised covariance ($\boldsymbol{\Sigma} + \varepsilon\mathbf{I}$, $\varepsilon{=}10^{-4}$, pseudoinverse for collinear dimensions). Lee et al. [19] define a confidence score over the last hidden-layer representation of a neural network (we write $\psi(\mathbf{x})$ for their feature extractor to avoid collision with both the CBM classifier $f{=}h \circ g$ of Eq. 1 and the NeSy MLP parameters ϕ), where high values indicate in-distribution samples:

$$M(\mathbf{x}) = \max_c \big[-(\psi(\mathbf{x}) - \boldsymbol{\mu}_c)^\top \hat{\boldsymbol{\Sigma}}^{-1} (\psi(\mathbf{x}) - \boldsymbol{\mu}_c)\big].$$

We apply the same framework but substitute the concept activation vector $\hat{\mathbf{c}} = g(\mathbf{x})$ for $\psi(\mathbf{x})$, since the concept bottleneck is the learned representation space in our architecture. Negating M (using $\max_c[-D_c^2] = -\min_c[D_c^2]$) then gives an anomaly score where high values indicate OOD samples:

$$s(\mathbf{x}) = -M(\mathbf{x})\big|_{\psi(\mathbf{x}) \to \hat{\mathbf{c}}} = \min_c\,(\hat{\mathbf{c}} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\hat{\mathbf{c}} - \boldsymbol{\mu}_c). \tag{3}$$

This polarity aligns directly with AUROC computation where OOD is the positive class [21].

Algorithm 1 summarises the full training and scoring procedure.

Algorithm 1 CBM Training and OOD Scoring

Require: Training set $\{(\mathbf{x}_i, \mathbf{c}_i, y_i)\}$, hyperparameters λ, γ
Ensure: Encoder g, classifier h, per-class Gaussians $\{\boldsymbol{\mu}_c, \boldsymbol{\Sigma}\}$
Initialise concept encoder g (2-layer MLP), linear classifier h
for each epoch **do**
  for each mini-batch $(\mathbf{X}, \mathbf{C}, \mathbf{Y})$ **do**
    $\hat{\mathbf{C}} \leftarrow g(\mathbf{X})$ ▷ concept scores $\in [0,1]^K$
    $\hat{\mathbf{Y}} \leftarrow h(\hat{\mathbf{C}})$ ▷ class logits
    $\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{CE}}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda\,\mathcal{L}_{\mathrm{BCE}}(\hat{\mathbf{C}}, \mathbf{C}) + \gamma\,\|\hat{\mathbf{C}} - \mathbf{C}\|^2$
    Update g, h via $\nabla\mathcal{L}$
  end for
end for
for each class c **do** ▷ fit OOD detector on training concepts
  $\boldsymbol{\mu}_c \leftarrow \mathrm{mean}\{g(\mathbf{x}_i) : y_i = c\}$
end for
$\boldsymbol{\Sigma} \leftarrow$ regularised shared covariance

Inference($\mathbf{x}$):
$\hat{\mathbf{c}} \leftarrow g(\mathbf{x})$; $\hat{y} \leftarrow \arg\max\, h(\hat{\mathbf{c}})$
$s(\mathbf{x}) \leftarrow \min_c\,(\hat{\mathbf{c}} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\hat{\mathbf{c}} - \boldsymbol{\mu}_c)$ ▷ OOD score
return $\hat{y}$, $s(\mathbf{x})$

C. Neuro-Symbolic NIDS: Differentiable Rule Learning

1) Rule Templates: $R{=}8$ rule templates per dataset are predefined from domain knowledge of IoT attack traffic; they are not generated by the model but provided a priori. Each template specifies a set of feature indices, condition directions ($>$ or $<$ a threshold), and initial threshold values. CTU-IoT-23 rules are listed in Table I; CIC-IoT-2023 rules appear in the right half of the same table.

Given input $\mathbf{x}$, each rule r is specified from domain knowledge as a conjunction (logical AND) of threshold conditions over a subset of features, with $\theta_i \in \mathbb{R}$ as initial threshold values. Each single condition maps to an indicator function,

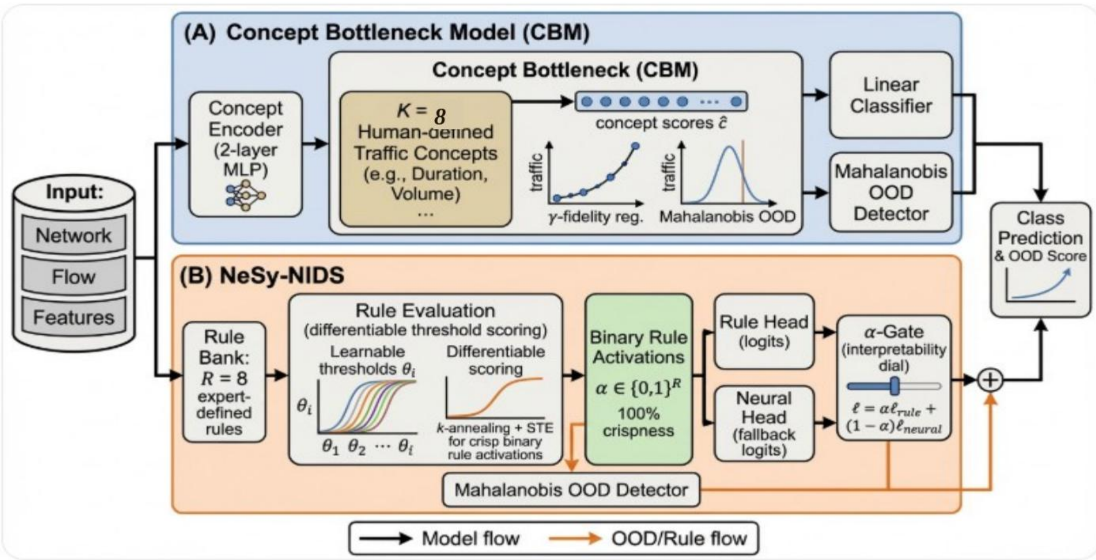


Fig. 1: System architectures. **(A) Concept Bottleneck Model (CBM):** A 2-layer MLP concept encoder transforms raw features into K interpretable concept scores; a linear classifier maps concept scores to class logits. Predictions are fully auditable at the concept layer, and expert intervention can override individual concept values at test time. **(B) NeSy-NIDS:** Expert-defined threshold rules are differentiably parameterised; k-annealing and the STE produce strictly binary rule activations at inference. A learnable α-gate blends the rule head with a parallel neural fallback.

e.g. $\mathbf{1}[x_i{>}\theta_i] \in \{0,1\}$, which equals 1 if the condition holds and 0 otherwise. Logical AND of two conditions is then the product of their indicators, since $\mathbf{1}[A] \cdot \mathbf{1}[B] = 1$ if and only if both A and B are true (and 0 otherwise). Extending this to all conditions in rule r, where $\mathcal{I}_r^{>}$ ($\mathcal{I}_r^{<}$) indexes features with greater-than (less-than) conditions:

$$a_r^{\text{hard}}(\mathbf{x}) = \prod_{i \in \mathcal{I}_r^{>}} \mathbf{1}[x_i{>}\theta_i] \cdot \prod_{j \in \mathcal{I}_r^{<}} \mathbf{1}[x_j{<}\theta_j].$$

For example, the `incompl_handshake` rule (Table I) targets C&C heartbeat traffic and has $\mathcal{I}_r^{>}{=}\{\text{syn}\}$ and $\mathcal{I}_r^{<}{=}\{\text{ack}\}$, giving $a_r^{\text{hard}}{=}\mathbf{1}[\text{syn}{>}2] \cdot \mathbf{1}[\text{ack}{<}1]$. This equals 1 only when both conditions hold simultaneously: an elevated SYN count (many connection initiations) *and* a suppressed ACK count (few completions). Neither condition alone is sufficient—high SYN traffic can arise in legitimate servers, and low ACKs appear in normal UDP flows—but their conjunction uniquely identifies half-open connections characteristic of C&C heartbeats. The product enforces this joint requirement: if either indicator is 0 (e.g., ACKs are present, meaning connections complete normally), the entire product collapses to 0 and the rule does not fire. This rule is non-differentiable, preventing gradient-based refinement of θ_i. To enable training, we adopt the differentiable logic principle of Petersen et al. [15]: the real-valued relaxation of logical AND over soft binary inputs $a_i \in [0,1]$ is their product, $f_{\text{AND}}(a_1, a_2) = a_1{\cdot}a_2$ (Table 1 therein). Since our inputs are continuous network traffic features rather than pre-binarized values, we first convert each condition to a soft binary using a sigmoid gate: $\mathbf{1}[x_i{>}\theta_i] \to \sigma\big(k(x_i{-}\theta_i)\big) \in [0,1]$, with the sign flipped for less-than conditions: $\mathbf{1}[x_j{<}\theta_j] \to \sigma\big(k(\theta_j{-}x_j)\big)$. Applying AND-as-product to these soft values gives:

$$a_r(\mathbf{x}; k) = \prod_{i \in \mathcal{I}_r^{>}} \sigma\big(k(x_i{-}\theta_i)\big) \cdot \prod_{j \in \mathcal{I}_r^{<}} \sigma\big(k(\theta_j{-}x_j)\big), \quad (4)$$

which recovers a_r^{hard} in the limit $k \to \infty$ [16]. Because θ_i are optimised jointly with the classifier, the model refines expert-specified rule boundaries to better match the observed data distribution.

2) k-Annealing and Straight-Through Estimator: We need the sigmoid steepness k to serve two conflicting roles: early in training, k must stay small so that gradients flow through smooth, wide boundaries while rule thresholds $\boldsymbol{\theta}$ are still being learned; later, k must be large so that each sigmoid collapses to a near-step-function and rule activations become effectively binary. We satisfy both requirements with a linear warmup schedule of our own design—the same interpolation pattern used for learning-rate warmup in transformer training, applied

TABLE I: NeSy-NIDS Rule Templates for CTU-IoT-23 (left) and CIC-IoT-2023 (right; features z-scored). Initial thresholds are derived from domain knowledge; θ values are updated during training.

CTU-IoT-23			CIC-IoT-2023		
Rule	**Conditions**	**Target**	**Rule**	**Conditions**	**Target**
`ddos_high_rate`	Rate>1000	DDoS	`syn_flood_ddos`	rate>0.15, syn>1.5	DDoS-SYN
`cc_hb_beacon`	1.5<IAT< 5, 0.1<dur< 5	C&C-HB	`udp_mirai_flood`	udp< -1.0, rate>0	Mirai
`incompl_handshake`	syn>2, ack<1	C&C-HB	`recon_port_scan`	fwd_iat>1.5	Recon
`okiru_low_rate`	Rate<500, IAT>2.5	Okiru	`vuln_scan`	pkt>1.0, rate>0	Scan
`asymmetric_flood`	src<1, dst>1	DDoS	`benign_low_rate`	rate<0, syn<0	Benign
`persistent_conn`	IAT>2.5	Benign	`large_pld_flood`	pld>0.4, rate>0	Flood
`benign_probe`	pkt<2, bytes<100	Benign	`hi_entropy_scan`	ent>0.5, iat>0.5	Scan
`ddos_short_burst`	Rate>1000, IAT<0.1	DDoS	`short_conn_flood`	dur< -0.003, rate>0.1	Flood

here to the steepness parameter. Concretely, k starts at $k_0{=}1$ (smooth sigmoid) and grows linearly to $k_T{=}10$ (near-step-function) over the first $T_{\text{warm}}{=}30$ training epochs, then is held fixed. At training epoch t this gives:

$$k(t) = k_0 + \min\!\big(\tfrac{t}{T_{\text{warm}}}, 1\big)(k_T - k_0), \tag{5}$$

where k_0 is the initial steepness, k_T is the target steepness, T_{warm} is the warmup duration in epochs, and t is the current epoch. The $\min(\cdot, 1)$ cap ensures k does not exceed k_T once warmup ends: at $t{=}0$ the fraction is 0 so $k{=}k_0$; at $t{=}T_{\text{warm}}$ it is 1 so $k{=}k_T$; for all later epochs it is clamped to 1 and k stays at k_T. At $k{=}10$ the sigmoid resembles a step function, but gradients remain nonzero. To enforce exact binary activations at inference while preserving differentiable threshold learning during training, we apply the Straight-Through Estimator (STE) [6], [7]. Bengio et al. [6] establish the STE principle: use the hard (binarized) value in the forward pass and pass gradients through the smooth surrogate in the backward pass. We apply this to $a_r(\mathbf{x}; k(t))$ from Eq. 4:

Forward pass. Binarize at the natural threshold of 0.5 (since $a_r \in [0, 1]$):

$$a_r^{\text{hard}} = \mathbf{1}[a_r(\mathbf{x};\, k(t)) > 0.5]. \tag{6}$$

Backward pass. The STE substitutes $\partial\mathcal{L}/\partial a_r^{\text{hard}}$ with the gradient of the smooth surrogate a_r. Since $\partial\sigma(u)/\partial u = \sigma(u)(1{-}\sigma(u))$ and θ_i enters $g_i^{>}{=}\sigma\big(k(t)(x_i{-}\theta_i)\big)$ with a minus sign, the sigmoid derivative with respect to θ_i is $\partial g_i^{>}/\partial\theta_i = -k(t)\, g_i^{>}(1{-}g_i^{>})$. Letting g_j denote any other sigmoid gate in rule r (of either direction), the product rule applied to Eq. 4 gives:

$$\frac{\partial\mathcal{L}}{\partial\theta_i} \approx \frac{\partial\mathcal{L}}{\partial a_r} \cdot \Big(-k(t)\, g_i^{>}(1{-}g_i^{>}) \prod_{j\neq i} g_j\Big), \quad i \in \mathcal{I}_r^{>}. \tag{7}$$

For $j \in \mathcal{I}_r^{<}$ the sign flips to $+k(t)$ because $g_j^{<}{=}\sigma\big(k(t)(\theta_j{-}x_j)\big)$ and $\partial g_j^{<}/\partial\theta_j{=} + k(t)\, g_j^{<}(1{-}g_j^{<})$. As $k(t)$ grows, the term $g_i^{>}(1{-}g_i^{>})$ naturally approaches zero once a gate commits to $\{0, 1\}$, reducing spurious threshold drift in already-decided conditions. This achieves 100% *rule crispness*—the fraction of activations in $[0, 0.1) \cup (0.9, 1]$ at inference—while preserving differentiable threshold learning during training.

3) Architecture and α-Gate: A Rule Bank layer stacks all R rule activations into a hard binary vector $\mathbf{a}(\mathbf{x}) \in \{0, 1\}^R$ at inference (or soft $[0, 1]^R$ during training). A linear head maps activations to rule logits $\mathbf{l}_{\text{rule}}{=}\mathbf{W}\mathbf{a}{+}\mathbf{b}$. A parallel two-layer MLP on raw features produces neural logits $\mathbf{l}_{\text{neural}}$. A scalar mixing gate blends the two:

$$\mathbf{l} = \alpha\, \mathbf{l}_{\text{rule}} + (1{-}\alpha)\, \mathbf{l}_{\text{neural}}, \tag{8}$$

where $\alpha{=}\sigma(w_\alpha) \in (0, 1)$ and w_α is a scalar parameter learned jointly with the rest of the network. The sigmoid constrains α to $(0, 1)$, guaranteeing a valid convex mixture while keeping the gate differentiable end-to-end. To push α toward 1 and favour rule-based predictions, we optionally add a squared penalty on the neural contribution $(1{-}\alpha)$:

$$\mathcal{L}_\alpha = \mathcal{L}_{\text{CE}}(\mathbf{l}) + \lambda_\alpha(1{-}\alpha)^2, \tag{9}$$

where $\mathcal{L}_{\text{CE}}(\mathbf{l})$ is the cross-entropy loss on the blended logits $\mathbf{l}$ from Eq. 8, and $\lambda_\alpha{\geq}0$ is a scalar hyperparameter controlling regularisation strength ($\lambda_\alpha{=}0$ recovers plain cross-entropy). The squared form is chosen because it is smooth and differentiable at the optimum $\alpha{=}1$, so gradient descent converges without subgradient approximations.

4) OOD Scoring in Rule-Activation Space: At inference, binary rule activations $\mathbf{a}^{\text{hard}}(\mathbf{x})$ form an R-dimensional embedding. We fit the Mahalanobis detector (Eq. 3) in this space using known-class training activations. Novel attacks are expected to activate different rule subsets from known classes, producing anomalous patterns detectable by distance.

Algorithm 2 summarises the full NeSy-NIDS training and scoring procedure, where $\text{sg}(\cdot)$ denotes the stop-gradient operator: its argument is passed through unchanged in the forward pass but contributes zero gradient in the backward pass, so only the smooth surrogate $\mathbf{a}$ receives gradient during the STE step.

D. Baselines and Evaluation

An unconstrained two-hidden-layer MLP (256, 128 units, ReLU, dropout 0.3) serves as the accuracy ceiling. A Decision Tree (DT; depth search over $\{5, 10, 15, \infty\}$, selected by validation F1) and a Random Forest (RF [26]; 100 trees, depth 10) represent classical baselines. DT and RF OOD scoring uses Mahalanobis distance in the raw feature space.

Algorithm 2 NeSy-NIDS Training and OOD Scoring

Require: Training set $\{(\mathbf{x}_i, y_i)\}$, rule templates with initial thresholds $\boldsymbol{\theta}^{(0)}$, hyperparameters λ_α, $k_0{=}1$, $k_T{=}10$, T_{warm}
Ensure: Thresholds $\boldsymbol{\theta}$, MLP ϕ, gate w_α, per-class Gaussians $\{\boldsymbol{\mu}_c, \boldsymbol{\Sigma}\}$

Initialise $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}^{(0)}$, MLP parameters ϕ, gate w_α
for epoch $t = 1, \ldots, T$ **do**
  $k \leftarrow k_0 + \min\big(t/T_{\text{warm}}, 1\big)(k_T - k_0)$ ▷ anneal steepness
  for each mini-batch $(\mathbf{X}, \mathbf{Y})$ **do**
    for each rule r **do**
      $a_r^{>} \leftarrow \prod_{i \in \mathcal{I}_r^{>}} \sigma(k(x_i - \theta_i))$
      $a_r \leftarrow a_r^{>} \cdot \prod_{j \in \mathcal{I}_r^{<}} \sigma(k(\theta_j - x_j))$
    end for
    $\mathbf{a}^{\text{hard}} \leftarrow \mathbf{a} + \text{sg}\big(\mathbf{1}[\mathbf{a}{>}0.5] - \mathbf{a}\big)$ ▷ STE binarise
    $\mathbf{l}_{\text{rule}} \leftarrow \mathbf{W}\mathbf{a}^{\text{hard}} + \mathbf{b}$
    $\mathbf{l}_{\text{neural}} \leftarrow \text{MLP}_\phi(\mathbf{x})$
    $\alpha \leftarrow \sigma(w_\alpha)$
    $\mathbf{l} \leftarrow \alpha\, \mathbf{l}_{\text{rule}} + (1{-}\alpha)\, \mathbf{l}_{\text{neural}}$
    $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}}(\mathbf{l}, \mathbf{Y}) + \lambda_\alpha (1{-}\alpha)^2$
    Update $\boldsymbol{\theta}$, ϕ, w_α via $\nabla \mathcal{L}$
  end for
end for
for each class c **do** ▷ fit OOD detector on binary rule activations
  $\boldsymbol{\mu}_c \leftarrow \text{mean}\{\mathbf{a}^{\text{hard}}(\mathbf{x}_i) : y_i = c\}$
end for
$\boldsymbol{\Sigma} \leftarrow$ regularised shared covariance

Inference$(\mathbf{x})$:
$\mathbf{a}^{\text{hard}} \leftarrow$ binarise rule activations at $k{=}k_T$
$\alpha \leftarrow \sigma(w_\alpha)$; $\hat{y} \leftarrow \arg\max\,(\alpha\, \mathbf{l}_{\text{rule}} + (1{-}\alpha)\, \mathbf{l}_{\text{neural}})$
$s(\mathbf{x}) \leftarrow \min_c\, (\mathbf{a}^{\text{hard}} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{a}^{\text{hard}} - \boldsymbol{\mu}_c)$ ▷ OOD score
return $\hat{y}$, $s(\mathbf{x})$

Metrics: weighted F1 on known-class test traffic; AUROC for OOD detection (unknown samples as positive); TPR at 5% FPR (TPR@5%FPR). For NeSy-NIDS: rule crispness and gate value α.

Training: Adam [27] (lr$=10^{-3}$, batch 512, up to 60 epochs, patience 12 on validation F1). Five independent seeds (0–4) for NeSy-NIDS; single run for all CBM variants.

IV. Results and Discussion

A. Classification Accuracy

Table II summarises all results. Both NeSy-NIDS and JointCBM (γ=0.5) match the unconstrained MLP on both datasets. The F1 gap between NeSy-NIDS and MLP is $+0.0013$ on CTU-IoT-23 and $+0.0032$ on CIC-IoT-2023—within one standard deviation. This validates the zero-accuracy-cost finding of [5] in the IoT NIDS domain under open-set evaluation conditions, and demonstrates that interpretable constraints carry no performance penalty.

The Rule-only ablation degrades substantially on CTU-IoT-23 (F1 = 0.5612), confirming that the neural fallback handles traffic patterns outside the eight rule templates and that the α-gate correctly delegates these cases. The fallback is less critical on CIC-IoT-2023 (F1 = 0.7561), where the tighter CIC rules cover more of the class structure.

Decision Tree (DT) and Random Forest (RF) achieve comparable F1 (DT: 0.9350/0.8116; RF: 0.9351/0.8045) but at severe cost to OOD detection, as detailed in Section IV-B.

Fig. 2 visualises weighted F1 across all models and both datasets. The shaded region groups the full γ ablation, making clear that γ has no effect on classification accuracy — the non-monotone behaviour identified in Section IV-F is purely an OOD phenomenon.

B. Open-Set Detection

Fig. 3 plots OOD AUROC against F1, revealing a clear separation between model families that depends on dataset structure.

CTU-IoT-23. Rule-space and concept-space Mahalanobis scoring are both highly effective. NeSy-NIDS achieves AUROC = 0.909 (σ = 0.017)—comparable to the unconstrained MLP baseline (0.913)—while JointCBM (γ=0.5) achieves 0.884. Both substantially outperform the DT (0.335) and RF (0.627), which apply Mahalanobis to the raw 37-dimensional feature space. The DT AUROC of 0.335 is below random (0.5), indicating that greedy classification splits produce representations that are *harder* to use for OOD separation than either the rule or concept space. The MLP baseline itself achieves AUROC = 0.913 via Mahalanobis in its learned 64-dimensional embedding, confirming that the interpretable bottleneck does not sacrifice OOD performance. The primary positive result is that interpretability does not reduce OOD detection; it is competitive with the unconstrained MLP and substantially improves it relative to classical rule-based models.

CIC-IoT-2023. The CIC results reveal a strong dependence of OOD detectability on the *degree of task coupling* in bottleneck training, not on compression per se. SequentialCBM—which learns concepts on binary targets before training the task head, without any gradient from the classification loss—achieves AUROC = 0.839, *exceeding* the DT baseline (0.809) despite compressing 39 raw features to the same 8-dimensional concept space. HybridCBM, which couples a subset of concepts to the task, achieves 0.808. By contrast, JointCBM degrades monotonically with stronger task coupling: from 0.627 (γ=1.0) down to 0.452 (γ=0.1), and NeSy-NIDS—whose binary rule thresholds are learned end-to-end against the classification objective—reaches only 0.613. The pattern is consistent: representations optimised jointly for task accuracy sacrifice the distributional spread required by Mahalanobis separation. We term this the *task-coupling OOD penalty*.

A secondary factor is feature-space coverage. On CIC, the DT assigns 34% importance to `syn_flag_number` and 33% to `number` (raw packet count)—both captured by the declared concept vocabulary (`is_syn_flood`, `is_high_rate`). Yet the DT also uses `https` (8%) and `time_to_live`

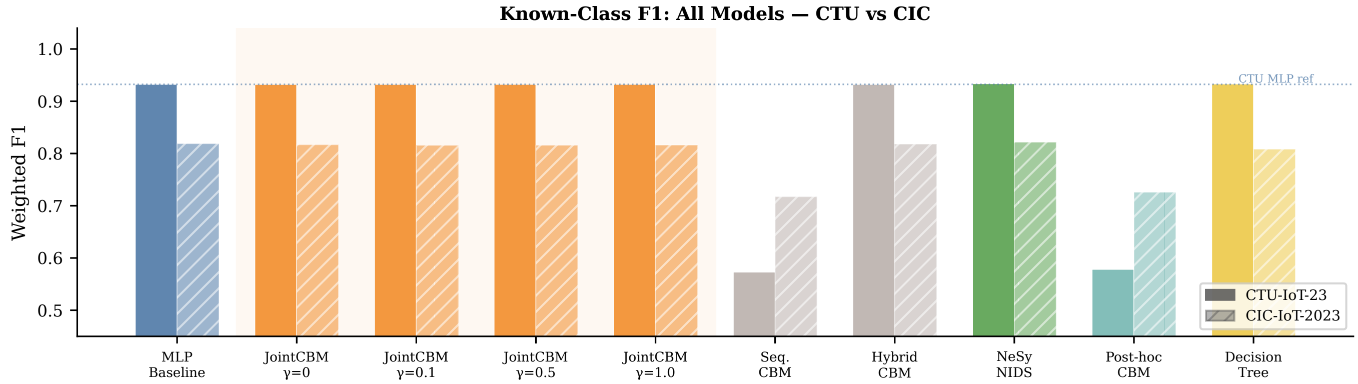


Fig. 2: Known-class weighted F1 for all models on CTU-IoT-23 (solid) and CIC-IoT-2023 (hatched). The shaded group covers the full $\gamma \in \{0, 0.1, 0.5, 1.0\}$ ablation; all four variants sit at the same F1 as the MLP baseline, confirming that γ-regularisation affects only OOD detection (see Table II). SequentialCBM and Post-hoc CBM incur large F1 penalties on CTU from decoupled bottleneck training.

TABLE II: Results on CTU-IoT-23 and CIC-IoT-2023. Mean $\pm$ std over 5 seeds where available; single run otherwise. AUROC uses Mahalanobis distance in the respective representation space (rule/concept activations for NeSy/CBM; raw features for DT/RF†). *Post-hoc CBM: linear concept probes on frozen MLP embeddings.

	CTU-IoT-23			CIC-IoT-2023		
Model	**F1**	**AUROC**	**TPR@5%**	**F1**	**AUROC**	**TPR@5%**
MLP Baseline	0.9323	0.913	0.596	0.8188	0.858	0.715
JointCBM (γ=0)	0.9322	0.812	0.426	0.8168	0.560	0.043
JointCBM (γ=0.1)	0.9323	0.795	0.414	0.8154	0.452	0.108
JointCBM (γ=0.5)	0.9325	0.884	0.687	0.8154	0.591	0.102
JointCBM (γ=1.0)	0.9324	0.864	0.605	0.8162	0.627	0.218
SequentialCBM	0.5731	0.678	0.386	0.7173	**0.839**	0.311
HybridCBM	0.9322	0.816	0.358	0.8181	0.808	0.183
Post-hoc CBM*	0.5783	0.864	0.490	0.7257	0.623	0.076
NeSy-NIDS	**0.9336**	$\mathbf{0.909 \pm 0.017}$	0.589 ± 0.101	**0.8220**	0.613	0.255
NeSy + α-reg	0.9336	0.895 ± 0.023	0.505	0.8205	0.634 ± 0.021	0.236
Rule-only	0.5612	0.863	0.459	0.7561	0.693	0.270
DT†	0.9350	0.335	0.037	0.8116	0.809	0.560
RF†	0.9351	0.627	0.411	0.8045	0.694	0.140

(2%) as discriminative splits—features absent from all eight concepts—preserving additional distributional signal that joint bottleneck training discards. For comparison, the CTU DT relies heavily on the Telnet port flag (35.3% importance), a class-specific artefact not captured by any concept and associated with its near-random CTU AUROC of 0.335. This asymmetry explains why the DT baseline is informative only on CIC: it happens to exploit features that fall outside the declared concept vocabulary, rather than being better in general. As demonstrated by Fort et al. [21], representation quality governs Mahalanobis OOD performance; SequentialCBM confirms that a concept bottleneck can *improve* OOD detection when training is decoupled from task pressure.

C. Rule Crispness and Class Selectivity

After k-annealing to k=10 and STE binarisation, all rule activations on both datasets fall strictly into $[0, 0.1) \cup (0.9, 1]$, confirming exact Boolean behaviour at inference. Rules either fire (value $\approx$1) or do not (value $\approx$0), with no residual soft logic.

Figs. 4a and 4b show the mean rule activation per class (class selectivity heatmaps). On CTU-IoT-23, `incomplete_handshake` fires at 0.859 for C&C-HeartBeat (SYN-heavy flows with few ACKs); `asymmetric_flood` activates at 0.796 for DDoS; and `cc_heartbeat_beaconing` at 0.772. These values are consistent with the rule definitions, validating that learned thresholds converge to meaningful boundaries.

Selectivity is extreme on CIC-IoT-2023:

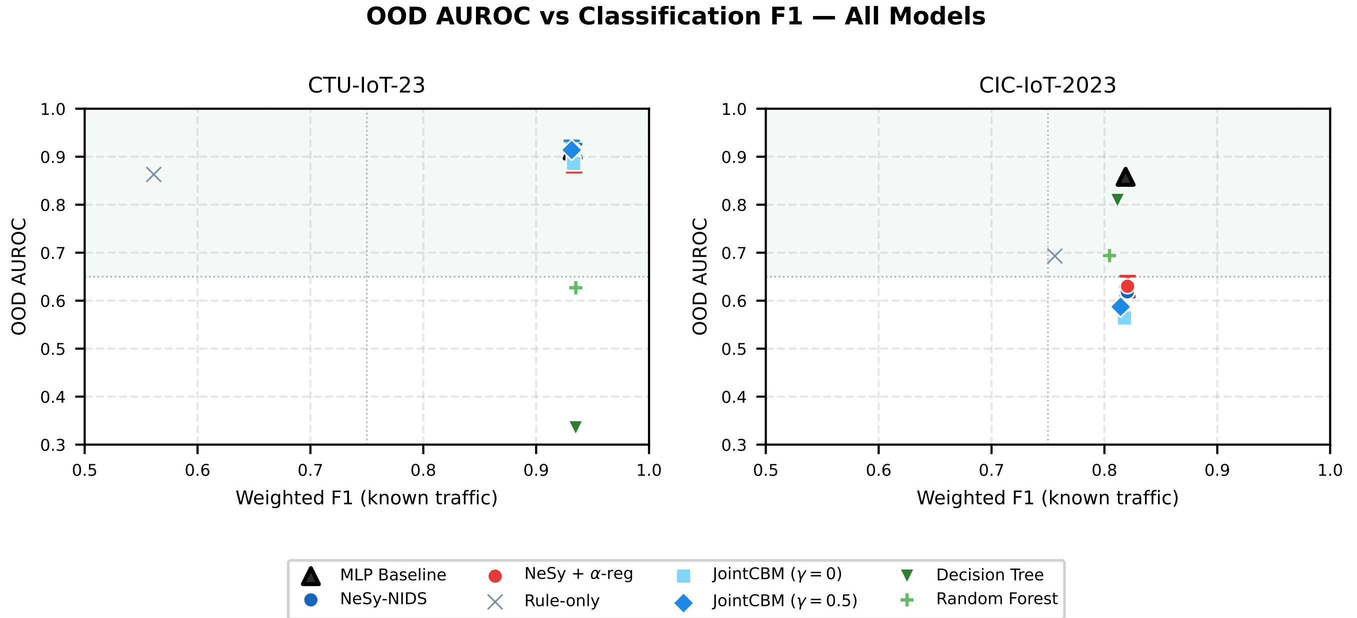


Fig. 3: OOD AUROC vs. known-class weighted F1 for all models on CTU-IoT-23 (left) and CIC-IoT-2023 (right). Error bars indicate ± 1 std over 5 seeds where available.

`syn_flood_ddos` activates at 0.996 for DDoS-SYN and `udp_mirai_flood` at 0.994 for Mirai. High selectivity explains both the strong classification performance and the OOD detection failure: rules that fire with near-certainty for specific known classes provide strong classification signal but little distributional spread for Mahalanobis separation.

D. Threshold Drift

Fig. 5 shows learned threshold drift (learned $-$ initial) per rule condition. On CTU-IoT-23, the IAT bounds of `cc_heartbeat_beaconing` shift by -0.2 to -0.4 (NeSy-NIDS learns a wider beaconing window); `asymmetric_flood` response-packet threshold shifts $+0.17$ (admitting more borderline asymmetric flows); and `incomplete_handshake` SYN threshold shifts $+0.19$ (tighter SYN requirement). All drifts are directionally consistent with domain knowledge, representing principled refinement of expert-specified rules rather than arbitrary deviation.

E. The α-Gate and Interpretability Trade-Off

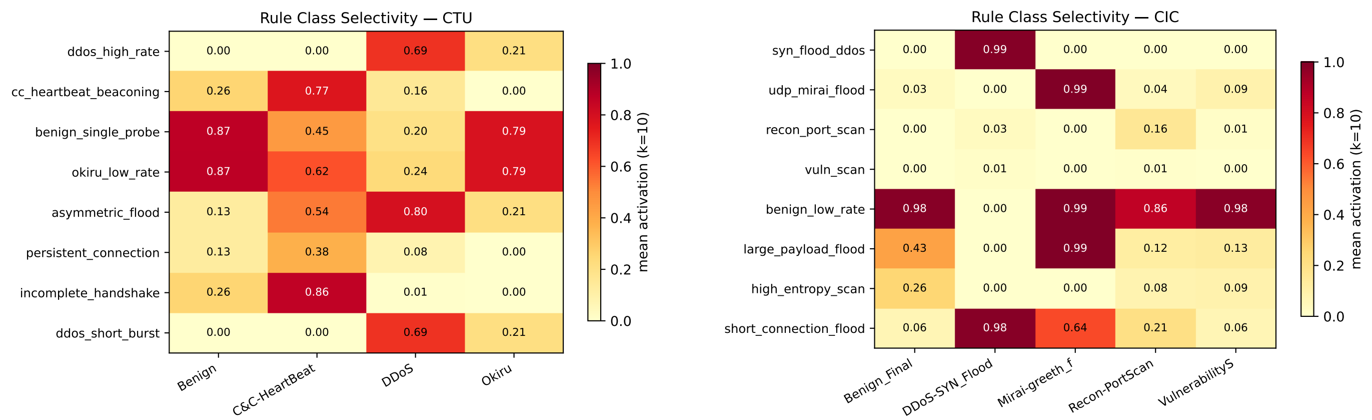
Without regularisation, NeSy-NIDS converges to α=0.37 on CTU-IoT-23 and α=0.27 on CIC-IoT-2023, indicating that the neural fallback carries substantial weight. With λ_α=0.1, the gate rises to α=0.68 and 0.85 respectively (seed 0), at zero F1 cost. On CTU-IoT-23, higher α incurs a minor OOD cost (AUROC: 0.909 $\rightarrow$ 0.895), consistent with the rule head displacing a neural head whose embedding was more OOD-compact. On CIC-IoT-2023, OOD AUROC marginally improves under α-regularisation (0.613 $\rightarrow$ 0.634), since the rule-dominant embedding is more tightly structured.

The α-gate provides a *deployable Pareto front*: an operator can select any interpretability-OOD trade-off point by adjusting λ_α during a brief fine-tuning step, without full retraining. This directly addresses the operator-control desideratum identified by Arp et al. [2].

F. Concept Leakage in CBMs

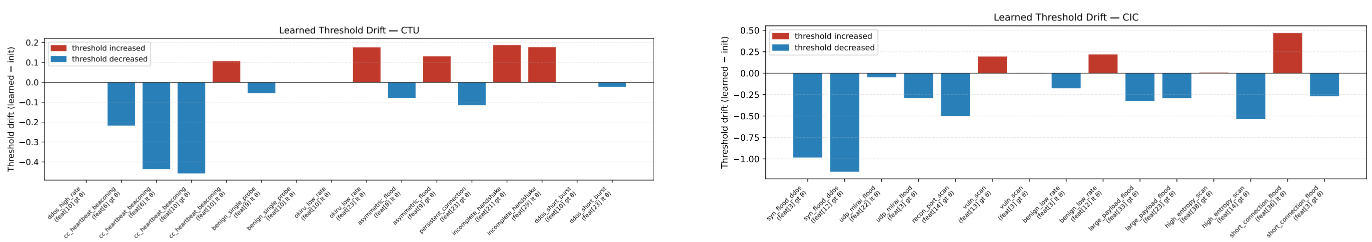
JointCBM trained without concept-fidelity regularisation (γ=0) achieves F1 = 0.9322 on CTU-IoT-23. Despite operating through an eight-concept bottleneck, the OOD AUROC rises from 0.812 (γ=0) to 0.884 (γ=0.5) when concept-fidelity regularisation is introduced, without any change in the bottleneck dimension or classification F1 (0.9325). This demonstrates that concept-fidelity regularisation compacts the concept space in a way that is directly beneficial to OOD detection: when concepts are forced to be semantically faithful, per-class clusters in concept space become more separated and Mahalanobis distances more discriminative. This is, to our knowledge, the first explicit quantification of concept leakage and its effect on OOD detection in the security domain, complementing the post-hoc fidelity analysis of xNIDS [3] and CADE [10].

Table II includes the full $\gamma \in \{0, 0.1, 0.5, 1.0\}$ ablation. The relationship between γ and OOD AUROC is non-monotone: γ=0.1 *worsens* OOD detection relative to γ=0 on both datasets (CTU: 0.812 $\rightarrow$ 0.795; CIC: 0.560 $\rightarrow$ 0.452). Partial regularisation at low γ disrupts the joint representation before concept predictions are sufficiently grounded, producing a concept space that is neither fully leaky nor semantically faithful. At γ=0.5 the grounding is sufficient to produce compact, well-separated per-class clusters in concept space, maximising



(a) CTU-IoT-23 (b) CIC-IoT-2023

Fig. 4: Rule class selectivity heatmaps. Cell values are mean rule activation per class at k=10 (post-STE). Darker cells indicate more selective activation for that class.



(a) CTU-IoT-23 (b) CIC-IoT-2023

Fig. 5: Learned threshold drift (learned − initial) per rule condition. Red (positive) indicates the threshold increased; blue (negative) indicates it decreased.

CTU AUROC. At γ=1.0 concept accuracy further improves (CTU: 0.927; CIC: 0.977) but CTU OOD slightly regresses ($0.884 \rightarrow 0.864$), suggesting marginal over-constraint of the concept bottleneck. On CIC, γ=0.5 and γ=1.0 converge (AUROC: 0.591 vs. 0.627), indicating that once task coupling is substantial, additional regularisation within the joint training regime yields diminishing OOD returns; decoupled training (SequentialCBM: 0.839) is required to escape the task-coupling OOD penalty entirely.

G. Computational Overhead

Table III reports parameter counts and per-sample inference latency (mean over 10,000 test samples, PyTorch CPU inference, single thread).

TABLE III: Parameter count and per-sample CPU inference latency.

Model	Parameters	Latency (μs/sample)
MLP Baseline	93,252	0.42
JointCBM	93,548	0.42
NeSy-NIDS	13,825	1.03

JointCBM adds K=8 concept predictions to the MLP with only 296 additional parameters (+0.3%) and no measurable inference overhead: both models process a sample in $0.42\,\mu$s. Concept-level interpretability is therefore computationally free relative to the unconstrained baseline. NeSy-NIDS is 7× more parameter-efficient (13,825 vs. 93,252) owing to the compact rule bank, but incurs a 2.4× latency overhead ($1.03\,\mu$s vs. $0.42\,\mu$s) from the product-of-sigmoids rule evaluation at each gate. Both models operate well within the memory and throughput constraints of a commodity IoT gateway or network tap.

H. Discussion

Dataset structure governs OOD detectability. CTU-IoT-23 has four well-separated known classes whose traffic profiles map cleanly within eight expert rules. CIC-IoT-2023's 29 unknown attack families largely share high-level characteristics (high packet rate, short connections) with known attacks, causing overlap in jointly trained rule and concept spaces. This aligns with Fort et al. [21]: representation quality governs Mahalanobis OOD performance. Our results refine this finding—SequentialCBM's decoupled concept space achieves AUROC = 0.839 despite the same 39→8 compression, demonstrating that the representation quality bottleneck is introduced by joint task optimisation, not by vocabulary size alone.

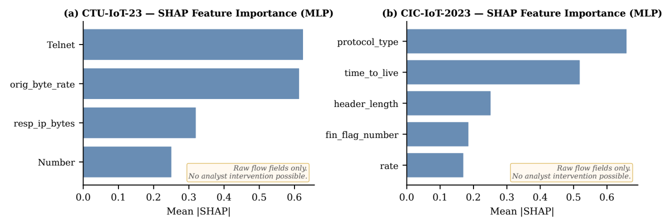


Fig. 6: SHAP mean |SHAP| feature importances for the MLP baseline on CTU-IoT-23 (left) and CIC-IoT-2023 (right). Top features are raw flow-level statistics (`Telnet`, `orig_byte_rate`, `protocol_type`, `time_to_live`) for which no analyst intervention is possible at inference time, in contrast to the directly overridable concept and rule activations of CBM and NeSy-NIDS.

Inherent vs. post-hoc interpretability. xNIDS [3] achieves high-fidelity post-hoc explanations from a black-box model. Our work demonstrates that inherently interpretable models match black-box accuracy while simultaneously supporting OOD detection in interpretable space, test-time concept intervention (CBM), and threshold auditing (NeSy-NIDS). The two approaches are complementary: deployments prioritising operator intervention would favour CBM, while those requiring forensic threshold inspection would favour NeSy-NIDS.

Fig. 6 illustrates the practical gap between post-hoc attribution and inherent interpretability. SHAP applied to the MLP baseline identifies `Telnet` and `orig_byte_rate` as top features on CTU-IoT-23, and `protocol_type` with `time_to_live` on CIC-IoT-2023 [1]. These are raw flow-level statistics that identify *which* features drive a prediction but offer no intervention mechanism—an analyst cannot correct a sample's `time_to_live` at inference time. CBM concepts (*is_high_rate*, *is_beaconing*) are explicitly binary predicates that an operator can read, verify, and override. NeSy-NIDS rules (`cc_hb_beacon`: $1.5 < \text{IAT} < 5$) are auditable threshold conditions with inspectable learned drift. Post-hoc attribution and inherent interpretability are therefore not equivalent: only the latter enables test-time expert intervention and forensic threshold auditing.

Interpretability evaluation. Interpretability is evaluated along three axes. *Structural*: 100% of NeSy-NIDS rule activations yield strict Boolean values after k-annealing and STE binarisation, confirming that inference is entirely rule-driven with no residual soft logic. *Semantic*: the class selectivity heatmaps (Figs. 4a and 4b) and threshold drift analysis (Fig. 5) confirm that learned rules remain aligned with domain knowledge—activations are selective for the classes an expert would expect, and threshold shifts are directionally consistent with the data distribution. *Fidelity*: for CBM, the OOD AUROC improvement from 0.812 to 0.884 as γ increases from 0 to 0.5—with F1 held constant at 0.932—directly quantifies how concept-fidelity regularisation compacts the concept space and improves OOD discrimination; γ thus provides a scalar fidelity handle. For NeSy-NIDS, the α-gate value (0.37 without, 0.68 with regularisation) directly quantifies the fraction of the classification decision attributed to the symbolic rule head and can be tuned at deployment without retraining. Collectively, these three axes provide a richer interpretability characterisation than any single scalar metric, and each is directly actionable: an operator can audit threshold drift to verify rule integrity, inspect selectivity heatmaps to confirm class attribution, and adjust α or γ to trade OOD performance for tighter concept or rule grounding.

Comparison to open-world NIDS. OWAD [4] targets normality shift detection in deployed anomaly detectors via hypothesis testing over time. Our setting addresses *unknown-class detection* at single-inference granularity in inherently interpretable models. The two approaches cover complementary phases of a deployed NIDS lifecycle: our method flags unknown-class traffic at inference time, while OWAD signals when the model's overall behaviour has drifted and retraining is needed.

Comparison to dedicated OOD scoring methods. Softmax-based OOD detectors such as MaxSoftmax [20], ODIN [28], and energy-based scoring [29] require a closed-set softmax classifier as their base model and produce OOD scores from output probabilities rather than representation geometry. OpenMax [30] fits per-class Weibull distributions in the penultimate layer of a standard network. All of these methods leave the model opaque: scores are derived from black-box representations that offer no human-auditable intermediate decision. By contrast, our Mahalanobis scoring operates natively in the concept or rule activation space—representations that are already interpretable by construction. A security analyst can trace an anomalous OOD score directly to which concepts or rules produced unusual activations, enabling actionable forensic inspection. Comparing against softmax-based detectors applied to opaque MLP representations would conflate OOD detection ability with interpretability cost; our baselines (DT, RF, MLP with raw-feature Mahalanobis) instead bound what is achievable without interpretable bottlenecks.

V. Conclusion

We presented a comparative empirical evaluation of two inherently interpretable architectures—CBMs and NeSy-NIDS—for open-set IoT NIDS on CTU-IoT-23 and CIC-IoT-2023. Both architectures match unconstrained MLP classification accuracy while delivering human-auditable intermediate representations. On CTU-IoT-23, both achieve competitive OOD AUROC (up to 0.909, comparable to the unconstrained MLP baseline of 0.913), substantially outperforming classical rule-based models. On CIC-IoT-2023, OOD detectability is governed by task-coupling strength, not compression: Sequential-CBM (decoupled training) achieves AUROC = 0.839, exceeding the DT baseline (0.809), while jointly trained variants degrade monotonically with coupling strength—a *task-coupling OOD penalty* that applies to bottleneck architectures trained end-to-end against the classification objective.

NeSy-NIDS contributes 100% crisp binary rule activations, semantically consistent threshold drift, and a controllable

α-gate. CBMs provide concept leakage quantification and elimination, with γ-regularisation simultaneously improving concept fidelity and OOD detection. Future work should explore expanding rule and concept vocabularies via data-driven template discovery, hierarchical architectures that combine concept-level and rule-level bottlenecks, and generalisation to enterprise-scale traffic where the ratio of known to unknown attack classes is less favourable.

Acknowledgment

Code and pre-trained models are available at https://github.com/crisphinen/interpretable-nids.

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